

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		<u>Earth's</u> magnetic field mentioned			1	1		1
	(b)		$\ln a = -3.22$ (1) 0.000 127 and -8.97 for B and $\ln B$ (1) Data point within $\frac{1}{2}$ square (1) ecf	1	1 1		3	3	3
	(c)		Method for gradient (1) Answer = -0.94 ± 0.01 (no penalty for missing minus) (1)	1	1		2	2	2
	(d)		Taking logs correctly e.g. $\ln B = \ln \left(\frac{\mu_0 I}{2\pi} \right) + \ln \left(\frac{1}{a} \right)$ (1) accept $\ln B = \ln \left(\frac{\mu_0 I}{2\pi a} \right)$ In correct form e.g. $\ln B = -\ln a + \ln \left(\frac{\mu_0 I}{2\pi} \right)$ (1)		2		2	1	2
	(e)		Any 3 × (1) from: -Straight line -Through all error bars or close to points or small scatter -Gradient close to -1 -Intercept -12 agrees with small permeability			3	3	3	3

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	(f)		At least one intercept read from the graph (-12.0 to -12.3) (1) Equating intercept to $\ln\left(\frac{\mu_0 I}{2\pi}\right)$ (1) A value of I obtained (23 A to 31 A) (1) Attempt at 2 values of I from 2 intercepts (regardless of mistakes) (1) $I = (27 \pm 4)$ A OR (26.7 ± 4.0) A ecf UNIT MARK and value for absolute uncertainty to 1 or 2 sig figs and consistent dps with I (1) Award a maximum of 4 marks for: Using a data point (from table or graph) (1) Getting a value for I (1) Getting 3-6 values of I (from table) or doing the max / min analysis on the data point (1) Expect $I = 25.2 \pm 1.2$ A UNIT MARK and value for absolute uncertainty to 1 or 2 sig figs and consistent dps with I (1) N.B. % method produces 1.2% which is far too small			5	5	3	5
	(g)		Data equally spaced for log (1) Data bunched (bottom left) in other graph (1)			2	2		2
			Question 5 total	2	5	11	18	12	18